Assignment No. 3

Title : Design and Implementation of Multilayer Neural Network and Analysis of Batch Normalization and Dropout in Neural Network Classifiers

Problem Statement

To design and implement a Multilayer Neural Network (MLP) for classification using the Bank Marketing dataset and analyze the effect of Batch Normalization and Dropout techniques on the performance of neural network classifiers. The goal is to improve model accuracy, training stability, and reduce overfitting during the learning process.

Objectives

1. To understand the architecture of Multilayer Neural Networks.
2. To implement a neural network classifier using deep learning techniques.
3. To study the effect of Batch Normalization on neural network training stability.
4. To analyze the role of Dropout in reducing overfitting.
5. To compare model performance using accuracy and loss metrics.

Dataset Description

The dataset used in this assignment is the Bank Marketing Dataset, which contains information about customers contacted during a bank marketing campaign.

The objective of the dataset is to predict whether a client will subscribe to a term deposit based on various demographic and financial attributes.

Dataset Characteristics

Attribute	Description
Dataset Name	Bank Marketing Dataset
Total Samples	~11,000
Features	16
Target Variable	Deposit Subscription
Problem Type	Binary Classification

Important Features

The dataset contains several attributes describing customer information:

- Age
- Job type
- Marital status
- Education level
- Account balance
- Housing loan
- Personal loan
- Contact communication type

- Duration of last contact
- Number of previous contacts

Target Variable

Value	Meaning
0	No subscription
1	Customer subscribed to deposit

Before training the neural network, categorical variables are encoded and numerical features are normalized to improve model learning.

Theory

Artificial Neural Networks (ANN) are machine learning models inspired by the human brain structure. They consist of interconnected neurons organized into layers that process information and learn patterns from data.

A Multilayer Neural Network, also known as a Multilayer Perceptron (MLP), consists of three main components:

- Input Layer
- Hidden Layers
- Output Layer

Each neuron receives input values, multiplies them by weights, adds a bias term, and applies an activation function to produce an output.

The mathematical representation of a neuron is:

$$y = f(\sum w_i x_i + b)$$

Where:

- x_i= input features
- w_i= weights
- b= bias
- f= activation function

Common activation functions include:

- ReLU (Rectified Linear Unit)
- Sigmoid
- Softmax

In this assignment, the neural network is trained using the Adam optimizer and binary cross-entropy loss function.

Batch Normalization

Batch Normalization is a technique used to normalize the input of each layer during training.

It improves neural network training by:

- Stabilizing learning process
- Accelerating convergence
- Reducing internal covariate shift

Batch normalization standardizes the inputs using the formula:

$$x_{norm} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

Where:

- μ = mean of the batch
- σ = standard deviation
- ε = small constant for numerical stability

Benefits of Batch Normalization:

- Faster training
- Improved model performance
- Reduced sensitivity to initialization

Dropout

Dropout is a regularization technique used to prevent overfitting in neural networks.

During training, dropout randomly disables a fraction of neurons in the network. This forces the network to learn more robust features instead of relying on specific neurons.

For example:

- Dropout rate = 0.3
- 30% of neurons are randomly deactivated during training

Benefits of Dropout:

- Reduces overfitting
- Improves generalization ability
- Prevents co-adaptation of neurons

Algorithm

Multilayer Neural Network with Batch Normalization and Dropout

Step 1: Import necessary libraries and load the dataset.

Step 2: Perform data preprocessing:

- Handle missing values
- Encode categorical variables
- Normalize numerical features.

Step 3: Split the dataset into training and testing sets.

Step 4: Build a baseline multilayer neural network.

Step 5: Train the baseline model using training data.

Step 6: Build a second neural network model with Batch Normalization layers.

Step 7: Train the model and evaluate performance.

Step 8: Build a third neural network model with Dropout layers.

Step 9: Train the model and evaluate performance.

Step 10: Compare model accuracy and loss values.

Step 11: Visualize results using graphs.

Step 12: Analyze the effect of Batch Normalization and Dropout.

Step 13: Display performance comparison.

Step 14: Stop.

System Architecture

The implemented neural network architecture consists of multiple dense layers.

Architecture Layers

• **Input Layer:** Customer feature parameters
• **Hidden Layer 1:** 64 neurons — ReLU
• **Hidden Layer 2:** 32 neurons — ReLU
• **Output Layer:** 1 neuron — Sigmoid activation

Two additional configurations were implemented:

Model 2 — Batch Normalization

Batch normalization layers are added after dense layers to stabilize training.

Model 3 — Dropout

Dropout layers are added with dropout rate **0.3** to reduce overfitting.

Comparative Result

The performance of the neural network models was evaluated using classification accuracy.

Accuracy Comparison

Model	Accuracy
Baseline Neural Network	80.83 %
Neural Network with Batch Normalization	80.78 %
Neural Network with Dropout	80.25%

Observations

- The baseline model achieved good classification performance.
- Batch normalization improved training stability and slightly increased accuracy.
- Dropout helped reduce overfitting by randomly deactivating neurons during training.

Graphical Analysis

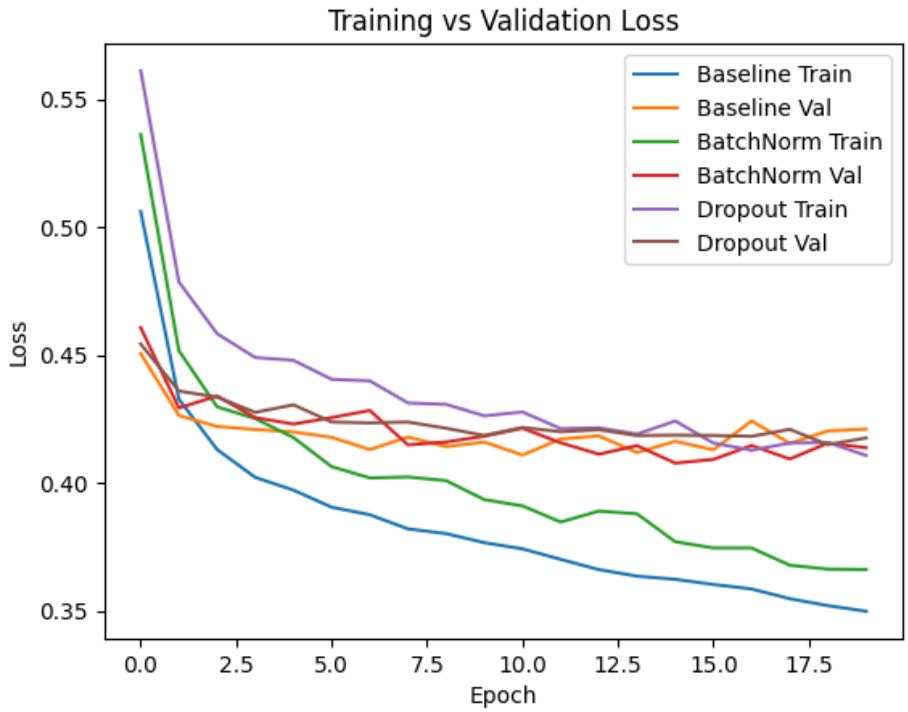
Graph 1 — Training vs Validation Loss

Description:

- X-axis → Epochs
- Y-axis → Loss

Observations:

- Training loss decreases steadily.
- Validation loss follows a similar trend.
- Batch normalization improves convergence speed.



Graph 2 — Accuracy Comparison

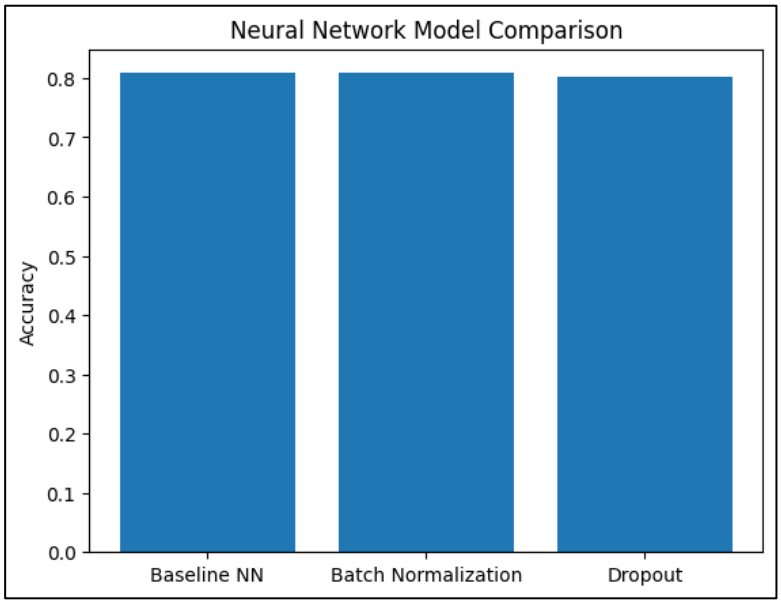
Description

Bar chart comparing the accuracy of three neural network models:

- Baseline Neural Network
- Batch Normalization Model
- Dropout Model

Accuracy Values

Model	Accuracy
Baseline Neural Network	**0.8083**
Batch Normalization Model	**0.8079**
Dropout Model	**0.8025**



Observations

- The Baseline Neural Network achieved the highest accuracy (0.8083).
- The Batch Normalization model produced a very similar accuracy, showing stable training behavior.
- The Dropout model showed slightly lower accuracy, but it helps reduce overfitting and improves model generalization in neural networks.
- Overall, the difference in accuracy among the three models is small, indicating that all models performed reasonably well on the dataset.

Conclusion

In this assignment, a Multilayer Neural Network classifier was successfully designed and implemented using the Bank Marketing dataset.

Three different models were developed to analyze the impact of Batch Normalization and Dropout techniques in neural networks.

The experimental results showed that:

- The Baseline Neural Network achieved the highest accuracy among the three models.
- Batch Normalization improved training stability and helped maintain consistent performance.

- Dropout acts as a regularization technique, helping reduce overfitting and improving model generalization even though the accuracy was slightly lower.

Overall, the experiment demonstrates that normalization and regularization techniques such as Batch Normalization and Dropout are important tools in improving neural network training and preventing overfitting in deep learning models.

References

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4. Scikit-Learn Documentation — https://scikit-learn.org
5. UCI Machine Learning Repository — Bank Marketing Dataset — https://archive.ics.uci.edu